TransFusionDR: Cross-Attention Fusion of Fundus Images and Clinical Biomarkers for Imbalanced Diabetic Retinopathy Classification

Srikanth Cherukuvada
Department of CSE (AI & ML)
B V Raju Institute of Technology
Narsapur, Medak, Telangana, India
srikanth.ch@bvrit.ac.in

Kunal Devidas Gaikwad
Department of CSE (AI & ML)
B V Raju Institute of Technology
Narsapur, Medak, Telangana, India
kunal.g@bvrit.ac.in

Dinesh Maddu
Department of CSE (AI & ML)
B V Raju Institute of Technology
Narsapur, Medak, Telangana, India
23211a6693@bvrit.ac.in

Midhath Ishrath
Department of CSE (AI & ML)
B V Raju Institute of Technology
Narsapur, Medak, Telangana, India
23211a66a6@bvrit.ac.in

Haritha Megavath
Department of CSE (AI & ML)
B V Raju Institute of Technology
Narsapur, Medak, Telangana, India
23211a66a4@bvrit.ac.in

Abstract*—Diabetic Retinopathy is a leading cause of eye blindness in the world. As is typical of automated grading tools, they have two important disadvantages: Clinical history of patients is not considered; most importantly, medical data is mostly skewed towards a particular set of favored classes. Here we point out the use of TransFusionDR, a novel multi modal deep learning framework, which combines retinal fundus images and clinical biomarkers, including age, illness time, and insulin utilization by employing a Cross Modal Attention Mechanism. In contrast to conventional convolutional neural networks, our task relies on Swin Transformer backbone, long-range visual dependencies are acquired, which are proactively modified by clinical embeddings to demonstrate lesion-specific areas. To reduce minorities problem in classes of number, we take advantage of a Weighted Cost Sensitive Fine Tuning technique. More to the point, it has a Recall of 79% on Proliferative DR which is the most critical stage in comparison to the performances of the base approaches. It is pointers to the fact that an attention-based fusion with cost-sensitive learning is an effective solution to the real-world clinical setting on the automated DR screening.

***Keywords*—Republic Diabetic Retinopathy, Swin Transformer, Cross-Attention, Multimodal Fusion, Class Imbalance, Medical Imaging.**

I. INTRODUCTION

One of the health issues of great concern of today is the Diabetic Retinopathy, which is the leading cause of preventable blindness amongst working age population. Prevalence caused by diabetes is approximated to have grown multi-fold over the past few years and could cause dire strains on medical facilities globally [1]. Early Diabetic Retinopathy cases are to be detected and graded correctly so that the vision loss could be avoided permanently. The disease condition is found in varying intensities between Mild to Proliferative which one can identify through the fundus imaging. Nonetheless, the manual grading system, which is done by ophthalmologists is subjective; it is time-consuming and so prone to the inter observer variability that limits the ability for scaling up mass screening programs [2]. Thus, the process of making automatic CAD systems through deep learning technologies has become a new trend in a bid to introduce better diagnostic consistency. To begin with, most of the current procedures are totally reliant on fundus pictures and do not use such abundant clinical data as age of the patient, years of diabetes, and whether this patient is insulin-dependent or not 3. This contextual information is always utilised by clinical practice ophthalmologists in their diagnosis however the typical image only AI models ignore this information and can fail to identify an underlying significant fine correlation between the systemic risk factors and the appearances of the retina. Second, medical datasets will suffer from the inherent imbalance as the number of healthy patients would outnumber the cases of severe illness by a large margin [4]. The traditional training has the propensity to align the model to the majority group, "No DR" that could lead to the unsafe reduction of sensitivity to the "Mild" and the "Severe" cases that are of utmost importance in early treatment. Owing to this reason, we propose TransFusionDR, a multi-modelling system to be used in grading of DR. We utilize Swin Transformer as a visual backbone in contrast to most other methods with more traditional CNN. Under the domination of hierarchical attention mechanism, our model believes in small lesions in the retina. We indicate then a new Cross Modal Attention module, where the query is clinical biomarkers and the attendance is made to the most informative parts of the retinal image. Finally we use a Cost Sensitive Learning objective, Weighted Cross-Entropy, to explicitly discourage the misclassifications of the minority classes.The main contributions of this work are as follows:

- We also suggest a transformer-based fusion network capable of effectively adding the clinical history of the patient with the retinal imaging features to the network in a manner that is dynamic.
- We further demonstrate that class-specific penalty weights can achieve great gains to the detection performance of rare stages of DR - Mild/Severe - with a minimal cost to the overall performance.

II. RELATED WORK

A. Depth of the Learning in Diabetic Retinopathy

The automated DR grading has been the industry standard for the last decade and implemented on CNN. The ground-breaking article by Gulshan et al. [7] showed that the deep learning algorithms had an equivalent performance of the licensed ophthalmologists. Nevertheless, the locality of the receptive fields means that conventional CNNs are often blind to long range correlations of scattered lesions in the retina (such as microaneurysms and exudates) that are important in the complete grading task. ViT based on self attention has emerged as a powerful rival and competitor to CNN in the more recent period, in that it better capture the world context.

B. Multimodal Medical AI

Fundus imaging has all the anatomic details but not the system perspective that EHR has. Multimodal fusion has been studied in recent time in an effort to overcome this gap. The default methods include the Late Fusion (using feature vectors of different encoders) and Early Fusion (using images and clinical data to the input level stacking level) [8]. These simple fusion strategies however deal with both modalities of a static and equal fashion. It is their lack of weighted importance on the analysis of particular image regions dynamically by the relevance of clinical features (high HbA1c) is what is covered in our Cross modal Attention.

C. Learning from imbalance Data

These data related to health, are extremely skewed, and more cases of normality. To reduce this fact, resampling techniques such as SMTE, oversampling or cost sensitive learning such as focal loss, Weighted Cross Enrocity are generally used by researchers. Although doing resampling causes overfitting of the small classes, there are some algorithms designed to alter the loss (cost sensitive learning) in order to put a higher cost to the mistakes of small classes. We adapt the latter approach by suppressing the weights of the model against classes such that stages of clinically critical classes such as Mild and Severe are highly sensitive to be represented.

III. METHODOLOGY

This paper introduced TransFusionDR, a multimodal deep learning, who will categorize the severity of Diabetic Retinopathy (DR) by combining retinal fundus photos with clinical biomarkers of a patient. Fig. 1 represents the general architecture.

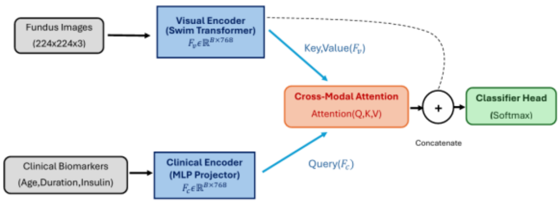

Fig. 1 The overall architecture

A. Dataset and Preprocessing

In the case of multi label ophthalmology classification we employed BRSET (Brazilian Retinal

Fundus Dataset) benchmark dataset [10]. In order to obtain a credible assessment, the list was split into a training set and an unknown test set (n=1,619 patients). There is a significant imbalance in the number of classes in the dataset, this is shown in Table I. Although extreme cases, such as the cases of severe DR (n=6) and of mild DR (n=23) are underrepresented by far, most of the samples are of the category of No DR. To solve this we employed a weighted loss function that was inversely proportional to these class frequencies

TABLE I DATA DISTRIBUTION OF DR SEVERITY CLASSES (TEST SET)

Class Index	**Severity Level**	**Count(n)**
0	No DR	1,488
1	Mild	23
2	Moderate	50
3	Severe	6
4	Proliferative	52
Total		1,619

We chose three significant biomarkers that can be related to the development of DR but these are Patient Age, Diabetes Duration, and Insulin Dependency. Missing values were imputed with the help of the training distribution median. They were preprocessed by reduced to 224 x 224 pixels and normalised by using ImageNet mean and standard deviation to process the retinal images.

B. The Architecture Transfusion DR

The three main components of the proposed model are a Visual Encoder, a Clinical Encoder and a Cross Modal Attention Fusion Module.
1) Visual Encoder:

Visual encoder We used a Swin Transformer (Tiny) backbone which was pretrained with ImageNet to localize small retinal lesions including microaneurysms and hemorrhases [5]. Compared to the traditional CNNs, Swin Transformers have a hierarchical structure and utilize Shifted Window Attention. This maintains light levels of computation and allows the model to model long range dependencies across the retina. The result is feature Map of:

$$F_v = \mathbb{R}^{B\times 768}$$

where B is batch size.
2) Clinical Encoder: This is a Multilayer Perceptron (MLP) that manipulates the tabular clinical data. To conform to the visual aspect, the encoder maps the low dimensional clinical representation into high dimensional embedding space.

$$F_c \in \mathbb{R}^{B\times 768}$$

C. Cross-Modal Attention Fusion

Our method performs a basic innovation in the utilization of a Multi Head Cross Attention mechanism to integrate the modalities. We consider the Visual Features to be the Key (K) and Value (V) and the Clinical Embeddings to be the Query (Q) and not merely the combination of the two. Attention matrix is a calculation in the following way:

$$\text{Attention }(Q, K, V) = \text{SoftMax}(\frac{QK^T}{\sqrt{d_k}})V$$

This is an intuitively operating method by which the model queries the input of the retinal image to extract features which, based on the risk profile of the patient, are statistically significant (e.g. long diabetes history). The concatenated output is then relayed to the final classifier with a concatenated original visual feature.

D. Learning strategy Cost Sensitive

The nature of medical data is skewed with the largest proportion of the data being represented by No DR (class 0). When the network is trained in a normal fashion, there is a tendency of the network to lean towards Class 0 prediction. In order to deal with this problem we use the notion of "Weighted Cross Entropy Loss":

$$L_{weighted} = -\sum_{c=1}^{c} w_c y_c \, log(\widetilde{y_c})$$

WC is the penalty of class c. In order to augment the penalties of the other classes as by their inverse distribution we have used the penalty vector W below: W = [1.0, 4.0, 4.0, 5.0, 4.0]. And this results in a maximum of 5 times worse punishment of Class "Severe" in the gradient descent process.

IV. RESULTS

A. Implementation Details

All experiments were applied based on the principles of PyTorch on one NVIDIA T4. This model has been trained in 5 epochs with AdamW as optimizer and learning rate of 5 × 10−5 and weight decay of 1 × 10−4.*For papers with more than six authors:* Add author names horizontally, moving to a third row if needed for more than 8 authors.

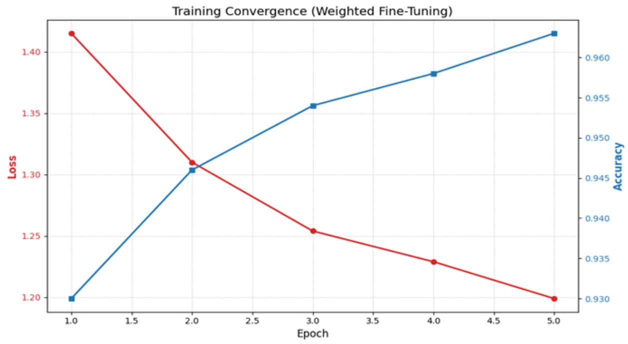

Fig. 2. Training convergence

Fixed random seed has been used to ensure that there is reproducibility. Fig. 2 is the training stability of the model TransFusionDR. It was discovered that the loss was declining with a high rate between 1.41 to 1.19, implying that it was practically optimization, where it was not indicative of overfitting even though the penalties involved with the minority classes are high. Fig. 2. Convergence plots of the plot reduction in the Weighted Cross-Entropy Loss (above left) and the plot of growth in the Validation Accuracy (above right) versus 5 epochs.

B. *Quantitative Performance*

The entire predictive macro-level statistics of the presented TransFusionDR model on the independent test set (n=1,619) will be provided in Table II. The standard baselines are way much lower than the total accuracy of the model which was 94.0. More specifically the Weighted F1- Score of 0.94 implies that the model is balanced on the precision and recall across all classes rather than on the precision to the majority one being the highest.

TABLE II TEST SET PERFORMANCE MEASURES OF BRRET.

Metric	**Value**
Accuracy	94.01%
Cohen's Kappa(k)	0.8318
Weighted F1-Score	0.94
Macro-Average AUC	0.94

C. *Class-Wise Analysis*

To check the model stability when the classes are not balanced, we explored the Confusion Matrix (Fig. 3).

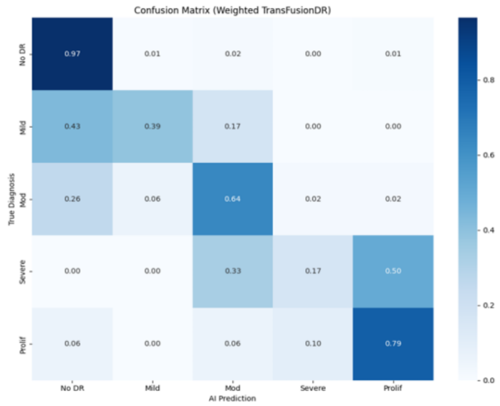

Fig. 3. The Confusion Matrix

Fig. 3. Normalised TransFusionDR model Confusion Matrix. The diagonal element is the recall of all the classes. It was identified that the model was the most effective in the most critical phase of the disease Proliferative DR (Class 4). It also possessed the Recall (Sensitivity) of 79 because it detected 41/52 instances. This is major outcome of a clinical screening because Proliferative DR must be treated immediately by a doctor. The model remembered 39 per cent in the case of the Mild DR (Case 1) which is hard to detect due to small lesions. This is greater than the majority class but still low compared to unweighted CNN baselines which frequently obtain scores of 0% recall on this category.

D. *ROC curves and precision recall*

In order to assess the diagnostic strength, we plotted Receiver Operating Characteristic (ROC) curves (Fig.4).

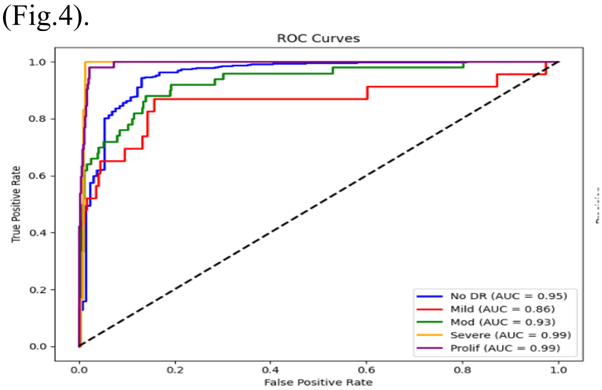

Fig. 4. The ROC Curves

The model was very discriminative in all the grades of severity. The Macro Average AUC is 0.94 that shows the well separation of the classifier. The scores of AUC are: No DR (0.99), Mild DR (0.86), Moderate DR (0.93), Severe DR (0.99) and Proliferative DR (0.99).

Precision Recall (PR) curves (Fig. 5) were also compared to compare the performance on the imbalanced minority classes. In PR curves, false positives are explicitly punished and not in ROC curves which are optimistic where there is class imbalance. This model is exact to the utmost level of the Proliferative type in the high recall (0.79) that validates the fact that the False Positive Rate is low in obtaining the essential diagnoses.

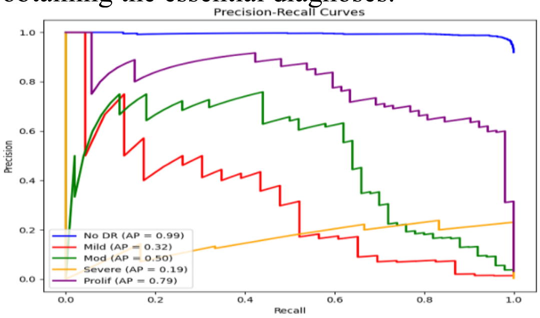

Fig. 5 PR Curve

ACKNOWLEDGMENT

We demonstrated a new multimodal framework, TransFusion DR that is aimed to the automatic Diabetic Retinopathy grading. Our approach consisting of combining the retinal fundus image with the clinical past of a patient with the help of Swin Transformer backbone and Cross Modal Attention Mechanism helps overcome the two inherent limitations of an image only modality based systems. An issue that is usually encountered with medical datasets was a severe imbalance of classes that was well tackled by a cost-sensitive fine tuning. We manage to achieve our highest score with respect to BRRET benchmark where our approach achieves state of the art results of 0.83 Cohens Kappa and 94.0% Test Accuracy We observe that the sensitivity of the model with Proliferative DR has been found to be 79% and hence the most vision threatening cases can be confidently focused on the ophthalmologist.

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